Sliding, swinging and rolling

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose the energy terms that match the type of motion.

SUCCESS CRITERIA - I can...

1. Distinguish translational and rotational kinetic energy.
2. Use equal height changes to compare ideal energy changes.
3. State when treating an object as a point mass leaves out important physics.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “translation” and “rotation” to explain your answer. Sentence starters are optional.

translation: motion of an object’s center from place to place

rotation: turning about an axis

pendulum: an object that swings about a fixed support

01 Warm-up: bring back an earlier idea

A **4 kg** object descends **6 m** near Earth.
1. Use g=**10**.
2. Find ΔU and the work done by gravity.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K_{translation} = \frac{1}{2}mv^2$; $K_{total} = K_{translation} + K_{rotation}$. A rolling object also rotates. Track all energy forms; an ideal pendulum exchanges gravitational and kinetic energy.

Worked reasoning: An object loses **20 J** of gravitational potential energy. If **8 J** becomes rotational kinetic energy and losses are negligible, only **12 J** becomes translational kinetic energy.

Your turn: Draw three energy bars and **explain why assigning all 20 J** to forward speed would be wrong.

Sentence starter: The step I would change is ... . I would change it because I found ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A slider and a rolling wheel start at the same height from rest. Why can the wheel reach the bottom with a lower center speed?

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A solid cylinder rolls without slipping from height **1.5 m**. Use $I=\frac{1}{2}mr^2$ and g=**10**. Derive v from $mgh=\frac{1}{2}mv^2+\frac{1}{2}I(v/r)^2$. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A pendulum's tension is perpendicular to its instantaneous motion in the ideal case where its length stays fixed. What work does tension do, and which interaction changes K?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

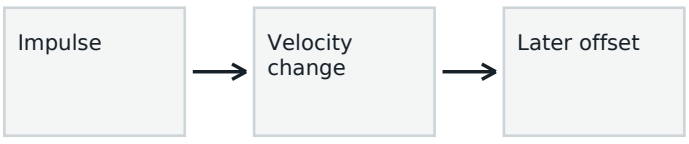
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Keep impulse fixed, multiply target mass by **4** and lead time by **6**. **Compare the modeled transverse shift** using an equation.
Label the quantities in the diagram and say what it treats as true.

Word help: lead time = time available before the modeled event

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3

07 Exit check

A point-mass pendulum drops **0.80 m** vertically from rest.
1. **Find speed** at bottom using g=**10** and neglect drag.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A thin hoop rolls without slipping from rest down the same **1.5 m** height as the solid cylinder in question **4**. Use $I = mr^2$ and g = **10 m/s²**. Derive its final center speed and compare with the cylinder.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3